

METROPOLITAN INTERNATIONAL UNIVERSITY
FACULTY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTING

**FORMAT OF RESEARCH REPORT (PROJECT REPORT) FOR THE DEGREES OF
BIT, BBC AND BCS
2019/2020**

General Format;

- a) The report is made up of **Six Chapters**, three of which are derived from the sections of your report, that is:
 - 1.0 INTRODUCTION,**
 - 2.0 LITERATURE REVIEW,**
 - 3.0 RESEARCH METHODOLOGY,**
 - 4.0 SYSTEM ANALYSIS AND DESIGN,**
 - 5.0 SYSTEMS DEVELOPMENT, and**
 - 6.0 CONCLUSION AND RECOMMENDATIONS**
- b) Although the proposal was in future tense (as should have been), **this PROJECT REPORT MUST BE PRESENTED IN PAST TENSE**. You cannot report about something in future or present. It is after studying a phenomenon that we report about it.
- c) The Team is expected to put at least 5 Preliminary Pages, with Page 1 being the Cover Page, and Page 2, being the Declaration Page, Page 3 being the Approval Page, Page 4 being Appreciation/ Dedication, and Page 5 having the Table of Contents. These will have Roman Numerals as numbers (Center Aligned).
- d) The Cover Page must NOT be numbered.
- e) The Declaration Page will include details of ALL members of the Group with their Official Names, Registration Numbers, and a Signature (As registered by Metropolitan International University); plus a space for a Date of Submission. *[For the purpose of the Final Dissertation (Project Report) this Particular Page will need to be scanned and inserted in the Document that will be uploaded on the E-Learning Platform].*
- f) Other preliminary pages may include an Abstract, Definition of Key terms, Abbreviations, Acknowledgements, List of Acronyms, and the like.
- g) The Supervisor has been advised to take attendance of the Weekly Meetings with the Teams, as may be convenient to him/her (Supervisor), such that members who have not been participating are identified and Scored/ Assessed accordingly.
- h) The Approval Page will have the Official Names of the Supervisor, and a Space for their Signature and Date of Signing.
- i) The Table of Contents **MUST be Inserted Automatically**, as you have been taught in MS Word classes. For more on this, please log onto

<https://support.office.com/engb/article/Create-a-table-of-contents-or-update-a-table-of-contents> and find online help.

- j) All References in Ref. List and Citations within text **MUST be** referenced in **APA Format, 6th Edition**. This Document can be accessed online, or download the specific attachment that will be sent together with this Document.
- k) Line Spacing for the contents of the Report must be 1.5lines (and not Double Space).
- l) The Team will use font Times New Roman, Black (Automatic), size 14 for Heading One, and font 12 for other headings and ALL other CONTENT as the standard font for the Project Report.
- m) The Document may have a maximum of Three Heading Levels per section, that is, INTRODUCTION will be Heading 1; the Background of the Study/ Project, Problem Statement, Research Goal/ General Objective/ Purpose of the Project, Research Objectives, Research Questions, formatted as Heading 2, and then any other Subsections under Headings 2, formatted as Heading 3.- Please refer to the Document/ Online Help about generating Automatic Table of Contents.
- n) The cover Page will be *Center Aligned*, while all other pages must be *Justify Aligned*.

Contents of the specific report sections are now detailed in the following pages.

METROPOLITAN



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**Topic [Do not Put the Word TOPIC HERE. Just state it, for example:] DEVELOPING A
COMPUTERIZED HUMAN RESOURCE INFORMATION SYSTEM FOR UGANDA
MEAT PRODUCERS COOPERATIVE UNION**

By

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[First Name first]

Department of Computing

**A Project Report Submitted to Metropolitan International University in Partial Fulfilment
for the Requirements for the Award of the Degree of Bachelor of Information
Technology of Metropolitan International University**

[1 line Space here]

Dec, 2019

[PRELIMINARY PAGE(S)]

DECLARATION

We, the undersigned, declare that to the best of our knowledge, this report is our original piece of work, and has never been published and/or submitted for any award in any other University or Higher Institution of Learning.

Dick Kasolo

Sign

Date:

APPROVAL

This report and project has been submitted with my approval as supervisor and my signature is here appended:

Signed.....

Date:

Mr. Moses Kasule

Metropolitan International University

DEDICATION

I would like to dedicate this piece of work to my mother, father and husband.

ACKNOWLEDGEMENT

I would like to thank my classmates, the staff of Metropolitan International University, and my parents who have educated me.

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LIST OF ABBREVIATIONS

ABSTRACT

CHAPTER ONE

INTRODUCTION

1.0 Introduction

Please give a brief about what is contained in this chapter.

1.1 Background to the Study

This section must be only 1 and half Pages. The Section may have 3 or 4 paragraphs (not indented). The Students should start by highlighting the Background of their Study Area [Topic], for instance in this case, we are studying Human Resource Information Systems. A funnel Approach is usually advised, where the writer may begin with a Global, or more General understanding of the Study Area, then narrow it down to national or Sectoral Scope, and then the value/ advantage/ rationale for using such systems [HRISs] if developed. And any other relevant data relating to the study area. The supervisor may advise another approach they are comfortable with. But the bottom line is, the team must highlight the study area. This paragraph must be appropriately referenced, and should *not necessarily* include definitions of the Topic. There must be logical flow from the first to the second paragraph. For example, the team may conclude this first paragraph by highlighting the business value that Ugandan firms may derive from developing HRIS. The Supervisor reserves the right to determine the ultimate content that appears here.

The students must then give a Background to the Organisation in the next/ second paragraph. They were proposing a new system for this particular organisation. This organisation must be one of the Field Attachment Organisations the students went to. The Team is advised to select the most appropriate organisation for their Research Interest. This should not be hard, as there are 5 people per group, with a maximum of Five Possible Organisations that can be studied. Information relating to location, when the organisation started, who stated it IS NOT NECESSARY. Please highlight the background to the Problem the Organisation was facing. In our example, Uganda Meat Cooperatives may have been facing issues relating to lost employee files and wrong postings of salaries. And the students here decide to design an automated HRIS to solve this. Please talk about the issues/ challenges that the organisation was facing before your study, how they came about, and how they were affecting the firm. This is a sort of Situational Analysis. What was wrong Out there? The Team is advised to cite any relevant Organisational Document to back-up their

assertions. This should then appear in the Appendices- if available. There should again be logical progression/ flow from this paragraph to the next.

The next section must summarize the Background, by highlight what the Team is going to do to help the organisation solve the challenges they may be facing.

1.2 Statement of the Problem

A maximum of half a page. This must be in three subsections combined into one paragraph. First highlight what the ideal situation is, then what is happening in the firm, and then the possible negative results if this continues to happen, and then what the team is planning to do to avoid these probable negative results. Another Approach may be given by the Supervisor. The Team **MUST** indicate evidence of the problem with appropriate citation.

Note: A persuasive problem statement consists of four parts: 1) the ideal, 2) the reality, 3) the consequences for the reader of the project report or the consequences for the organisation if the reality is not dealt with, and 4) the Project's proposed system and its advantage over the existing system.

Well-constructed problem statements will convince your audience that the problem is real and worth having you investigate/ design a system for. Your strategy is one of contrast: by talking about the ideal situation before talking about what the situation is as it exists, you can not only persuade the reader that a problem exists, but then go on to emphasize the consequences of ignoring or addressing the problem, and then proposing what you think is the solution you are going to design.

Remember, your problem statement is the backbone of the research and the project report. By giving careful consideration to how you construct it now (for the report), you can use it when doing your research (developing system), as well as the project report.

The Parts:

STATEMENT 1: (DESCRIPTION OF THE IDEAL SCENARIO- What should the situation be in the firm?) Describe the goals, desired state, or the values that **your audience considers important** and that are relevant to the problem.

(BUT)

Connect statements 1 and 2 using a term such as "but," "however,"

"Unfortunately," or "in spite of";

STATEMENT 2 (THE REALITY OF THE SITUATION)

Describe a condition that prevents the goal, state, or value discussed in statement 1 from being achieved or realized at the present time. What is wrong? What is on ground?

STATEMENT 3 (THE CONSEQUENCES FOR THE ORGANISATION if problem is not addressed)

Using specific details show how the situation in statement 2 contains little promise of improvement unless something is done. Then emphasize the benefits of research by projecting the consequences of possible solutions as well.

RESEARCH (YOUR PROPOSED RESEARCH TO INVESTIGATE THE POSSIBILITY OF MAKING THE REALITY MORE LIKE THE IDEAL.)

Describe the areas of inquiry you used that *led to* solutions to the problem--- how did you research the problem? What sources of information, types of research (primary or secondary), or tools did you use to help you find solutions and make recommendations to resolve the clash?

Example 1:

Example #1 STATEMENT 1.....the Ideal

In order to provide excellent online patient care at a minimal cost, Naguru KCCA Hospital needs an online diagnostic system that is accessible, safe, efficient, and accurate. In addition, the procedures should not be overly difficult for doctors to use.

STATEMENT 2.....the Gap.

Right now however, Naguru KCCA Hospital's main diagnostic tools are CAT scans and myelograms (spinal taps). The CAT scan (current system) fails to make clear diagnoses 60% of the time. When the CAT scan fails, doctors must resort to the myelogram. This procedure is very painful and sometimes dangerous for the patient.

STATEMENT 3

If Naguru KCCA Hospital continues to use the current diagnostic system, it will not only be wasting time and money, but put patients' health at risk. Undue suffering could lead patients to choose another hospital with more advanced facilities. This jeopardizes the overall efficiency and earning potential of the Hospital.

STATEMENT 4: Project Solution

A new diagnostic technique, online nuclear magnetic resonance (NMR), providing instant help from doctors around the world, may surpass the current CAT scan in accuracy and reduce the need to resort to the myelogram. This study proposes to design an online NMR lab at Naguru KCCA Hospital.

The: Students should combine these four sentences/ statements into one paragraph.

Example 2: The statements combined into One Paragraph as SHOUL BE.

“An automated Human Resource Information System can provide Human Resource professionals with opportunities to enhance their contribution to the strategic direction of

the firm, like reducing time wastage (Lawler *et al.*, 2013) and assisting in decision making (Hendrickson, 2003; Lengnick-Hall *et al.*, 2003). Uganda Meat Producers Cooperative Union currently has most of its Human Resource data processed manually, using files and paper to store employee's information, and manually processing the payroll. This according to the CEO leads to loss of data and delayed report production. If a new HR system is not adopted, UMPC risks exposing sensitive employee data to serious security breaches. This study is aimed at developing a Computerized Human Resource Information System for Uganda Meat Producers Cooperative Union Ltd which will help in data security, reduced paper work, interactive communication between employees and minimize manual data entry" (Mugumya 2014, pp.2).

1.3 Purpose of the Study/ General Objective

Example: This report sought to design and develop an automated HRIS for Uganda Meat Producers Cooperative Union Ltd to improve employee data management.

1.4 Specific Objectives of the Study [A maximum of Only 4, and a minimum of 3]

- a) To study and analyze the current Human Resource Information in Uganda Meat Producers Cooperative.
- b) To review literature in relation to human resource Information Systems management in organizations.- Or, To Identify Requirements for designing an HRIS for Uganda Meat Producers Cooperative.
- c) To design and develop the Human Resource Database Management System for Uganda Meat Producers Cooperative Union Ltd.
- d) To test the Human Resource Database Management System that has been developed.

1.5 Study Scope

1.5.1 Subject/ Conceptual Scope 1.5.2 Geographical Scope 1.5.3 Time Scope

1.6 Justification/significance of the Study These may highlight:

- a) Importance of the HRIS to the Organisation.
- b) Significance to the industry or other players.
- c) Significance to other researchers.

d) Significance to the Students.

Note:

- ✓ In total, this section must have **only 2 pages**.
- ✓ Sub-sections like *Research Questions*, *Hypotheses*, and *Conceptual/ Theoretical Framework*, which you have covered in your Research Methods theory classes **are NOT necessary** for both your Proposal and Reports.

CHAPTER TWO

LITERATURE REVIEW 2.0 Introduction

The review of Literature must be according to your objectives. Do not include data that does not help you meet the objectives. The team **MUST** not reproduce classroom notes here. Please read articles, books, and other reports relating to your study topic. Care must be taken not to plagiarize here. Read other people's work, get ideas, and write your own literature. Please reference all material here. All material must be paragraphed, without using bullets or lists. Please refer to guidance about APA referencing for citations.

In our example, the literature will be about:

2.1 Human Resource Information Systems 2.2 Applications of Human Resource Information Systems 2.3 Designing Human Resource Information Systems
2.3.1 Implementation of Human Resource Information Systems
2.4 Challenges in Designing and Implementing Human Resource Information Systems 2.5 Overcoming the Challenges 2.6 Conclusion

Note:

- ✓ This Section may include between 8-10 pages of Literature only.
- ✓ Appropriate referencing in text should be followed, as you have been guided in your Research methods classes. Please Use APA 6th Edition which has been attached.
- ✓ Only recent literature should be reviewed (preferably between 2010 and 2015), unless the group has used a known theory like TAM, TRA, Social Networks Theory or a known model. This (*the use of models or theories*) however, is not necessary at this level.
- ✓ The Supervisor has been advised to use Anti-plagiarism software to check for unreferenced work and any other grammatical errors. Please upload your document on Grammarly.com or any other freely available online anti-plagiarism tool that the Supervisor may recommend, and see that you don't have plagiarized material.

CHAPTER THREE

RESEARCH METHODOLOGY

3.0 Introduction

This section highlights the research methodology, methods, Research Design/ Research Approach, Population and Sample (where necessary), Sampling Technique (where necessary), Requirements Elicitation Methods/ Data Collection Methods, System Design Approaches, any Limitations of the Study, and ethical considerations if necessary.

3.1 Research Design/ Research Approach

The project team undertook a [*students can either choose Design Science Approach, Participatory Research Design, or Action Research Approach*]. Then the students will be expected to highlight what the chosen Research Design means, the steps involved in the approach/ design, and why they used it, and how this approach/ design helped the team to achieve the objectives. [For example, If Design Science is the chosen Approach, the students may say:

In this approach, the team designed an Information system artifact, through identifying business requirements and thereby defining a practicable/ pragmatic solution to the existing problem (Weber, 2010; Hevner et al., 2004; Au, 2001) in the organisation. The research team, then developed a working prototype of a [Human Resource] Information System that will improve [HR] management in the organisation.

Please use any other relevant approaches to writing this particular section, as may be guided by your Supervisor.

3.3 Study Population

According to the Company Payroll [*Give an authoritative document source to find the population*], the population at [*Your Organisation/ Research Area*] is [*number*] members. The target population was therefore comprise of all these members [*e.g. staff and top management*].

3.3.1 Sampling Design/ Sampling Technique [Optional]

[*Sampling may only be necessary if one is studying (designing a system for) a large organisation of employees more than 45, and the Project Team feels they need the views of a representative sample of all employees/ study group. Otherwise, this section is optional*].

Otherwise, the students may write something like this:

A sample of [number] members was randomly selected for the Study and Project. The sample size was determined according to Krejcie and Morgan, (1970). *[Please find table at <http://www.kenpro.org/sample-size-determination-using-krejcie-and-morgan-table/> if you need to use it]*. The researchers used (Simple Random/ Stratified/ Purposive/ Snowball) sampling to select the sample. This is because *[Sampling technique]* helps to *[give advantage of the sampling technique]*, e.g. This is because simple random sampling gives each member of the organisation an equal chance to participate in the project; *[or If Purposive Sampling is chosen]*. This is because with Purposive Sampling, only members of the IT department, who are relevant for the IS project were targeted, and this helped reduce the cost of data collection and requirements' elicitation.

3.4 Sources of Data

The researcher used both primary and secondary data. Primary data about the existing Information System was obtained from the Company documents, and interviews with members of the organisation/ study group, *[for example Managers, CEO, staff, library, manuals among others]*.

Secondary data was used from online articles and other literature available about HRIS.

3.5 Requirement Elicitation [Data collection] Techniques

The project team studied the existing system carefully and in detail. This helped the researchers to come up with a new system that enabled the organisation to improve the management of their [Human Resource] records. The researcher used observation, interviews, and *[Note that: participatory data collection techniques are encouraged here, like Mind Mapping, Affinity Diagrams, Locality Diagramming, Ishikawa Diagrams, Problem Trees, Focus Group Discussions]* to collect or elicit requirements for designing and developing the required information system. *[The students should ONLY use those techniques that are applicable and relevant for their project, as guided by the Supervisors]*.

3.6 System Analysis and Design

The students should mention here the systems design approach they used (either structured or object oriented design approach). They should then highlight why they chose one approach over the other, and how this may have helped in developing the information system.

3.7 System Design Approach

System design refers to designing and finally developing the system needed to solve the problem. The researchers must specify the design approach here. For example, Prototyping Method, Rapid Application Design, Extreme Programming, or generally the SDLC Approach. Students must specify why they are chose a particular development approach, and how it helped solve the problem/ achieve the Project Design Objectives/ Research Objectives.

3.8 Limitations of the Project

The group must give limitations or possible challenges that may have affected the Project or Study Process, and possible solutions to the limitations. These may be only 3, and indicated in either letters or roman numerals. E.g.;

- a) The Project was hindered by inability to get literature and data about the subject/ topic of study. The team plans used online sources to gather as much information about the study area as possible.
- b) The Project was affected by inability to get the latest software programmes to design the system. The team sourced for the most appropriate open Source software that can be used to solve the Problem.
- c) The team did not have enough programming or technical skills to develop the artifact. The team enrolled on online courses to improve their programming skills in order to design a High Quality software.
- d) The team experienced some degree of user resistance if the software/ system is taken up by the firm. The team however solved this by doing enough User Acceptance Training and Testing of the Program/ Information System before installation.

3.9 Ethical Considerations [Optional Sections which may be included].

These may be shown if human-sensitive/ Human Interaction programmes that manage sensitive data are being designed. The Research team may show us how they managed ethical usage of the systems, and generally research ethics here.

CHAPTER FOUR

SYSTEM ANALYSIS AND DESIGN

4.0 Introduction

4.1 Requirement Specification

4.1.1 Functional Requirements

4.1.2 Non-Functional Requirements

4.1.3 Systems Requirements

4.1.4 User Requirements

4.2 Systems Design

The researcher team is required to make a conceptual/theoretical framework for the system before developing it. A framework is a picture representation of a system. Systems design should follow proper design principle;

4.2.1 Structural/Procedural Design

The team may include the following sections:

- Users of the Database
- Data Dictionary; i.e. Entities with their attributes and data types
- Context Diagram
- Entity Relational Diagram (ERD)
- Data Flow Diagram (DFD) (From at least Level 0 to 1)
- Flowchart Diagram
- Hierarchical diagram
- Decision Table and Decision Tree
- Information/Data backup plan

4.2.1 Object Oriented Design

- Users of the Database
- Class Diagrams,
- Sequence Diagrams,
- Activity Diagrams,
- State Machine Charts,
- Use Case Modeling,
- Collaboration Diagrams,

- Package Diagrams,
- Component Diagrams,
- Deployment Diagrams etc

Or:

4.2 Integrated Web Solution or Mobile Apps Students may include the following sections.

- Users of the Website or Mobile App
- Conceptualisation – Goal, Structure, Navigation scheme
- Information design/ App Design
- E Business functionalities and integration (for integrated websites)
- Application Design features
- Database schema (for integrated websites)

4.3 System Testing

Before a project is put to use, it must be tested to see its compatibility with user requirements and how it effectively addresses the identified problems to identify bugs in the software. Please indicate the specific tests that you have undertaken for your new system. These may include:

- **Unit Test;** to test each module e.g. each form or webpage
- **Integration Testing ;** to test how well modules operate when linked as a single unit ▪
- **System test;** this tests the whole system to see how well it meets business requirements.
- **Volume testing,** to test whether the system can hold the intended amount of data.
- **Other tests** – validation testing, stress testing, alpha and beta testing (acceptance testing)

CHAPTER FIVE

SYSTEMS DEVELOPMENT

5.0 Introduction

Briefly describe what is included in this Chapter.

5.1 Systems Development

In brief:

Describe the new system in terms of input, processing, output, boundary, environment and feedback of the system.

5.2 System Implementation

This should be followed by some print screen pictures of the system.

The user guide should be given to explain how each screen (e.g. page, table etc) should be used. In other words, a future reader should be able to understand the system even without actually using it.

CHAPTER SIX

CONCLUSION AND RECOMMENDATIONS

6.0 Introduction

6.1 Conclusions

6.2 Recommendations

6.3 Limitations of the study

6.4 Future work

REFERENCES

- ✓ Must be in APA format as indicated in your RM classes ✓ Must have a Second-line indentation.
- ✓ Must not be numbered
- ✓ Must appear in Alphabetical Order
- ✓ Must not be too many (not more than 2 pages).
- ✓ Use & and not the word 'and' between names of more than one author. Please read RM notes.
- ✓ Pagination: Use the abbreviation p. or pp. to designate page numbers of articles from periodicals that do not use volume numbers, especially newspapers.
- ✓ Indentation*: The first line of the entry is flush with the left margin, and all subsequent lines are indented (5 to 7 spaces) to form a "hanging indent".
- ✓ Underlining vs. Italics*: It is appropriate to use italics instead of underlining for titles of books and journals.
- ✓ References to periodical articles must include the following elements: author(s), date of publication, article title, journal title (in italics), volume number, issue number (if applicable), and page numbers.

Read your RM notes for more on this.

Example: I will now use details of documents I have used (plus others) to write this document as our Reference list.

- Carr, A., & Kaynak, H. (2007). *Communication methods, information sharing, supplier development and performance: an empirical study of their relationships*. International Journal of Operations and Production Management, 27(4), 346-370.
- Chen, C., Baird, S., Ssentongo, K., Mehtsun, S., & Olapada-Olaopa, E.O. (2014). *Physician tracking in sub-Saharan Africa: Current initiatives and opportunities*. The George Washington University. Washington, DC; Health Sciences Research Commons. Retrieved from http://hsrc.himmelfarb.gwu.edu/sphhs_policy_facpubs
- Chung, S.H., Kelly, R.R., & Lewis, B.R. (2003). *The Impact of Information Technology Infrastructure Flexibility on Strategic Alignment and Applications Implementation*. Department of Computer Information Systems, Unpublished article

- Cruz, I. F., & Xiao, H. (2008). *A layered framework supporting personal information integration and application design for the semantic desktop*. The VLDB Journal 17:1385–1406, 17, 1385–1406. Doi: 10.1007/s00778-008-0102-5
- Curie, W. (2009). *Contextualising the IT Artifact: Towards a wider research agenda for IS using institutional theory*. Information Technology & People, 22(1), 63–77.
- Hevner, A., & Chatterjee, S. (2010). *Design Research in Information Systems*. Integrated Series in Information Systems 22, DOI 10.1007/978-1-4419-5653-8_2.
- Peffers, K., Tuunanen, T., Rothenberger, M.A., & Chatterjee, S. (2008). *A Design Science Research Methodology for Information Systems Research*. Journal of Management Information Systems, Volume 24 Issue 3, pp. 45-78.
- Weber, S. (2010). *Design Science Research: Paradigm or Approach?* AIS Electronic Library (AISeL).

APPENDICES

- ✓ Proposed Research Budget.
- ✓ Interview Guides/ Questionnaires [If used].
- ✓ Schedule of Activities/ Gantt chart.
- ✓ Company Payroll Sheet, [any Relevant Documents]

Note: The Supervisor will guide on the formats or these, and what is necessary to be included.